SOAR Playbook Development – Phishing Incident Response

PLAYBOOK

This playbook automates the response to phishing alerts. The SIEM detects suspicious activity, then the IP reputation is checked. If the IP is malicious, it is blocked using CrowdSec. Finally, a case is created in TheHive for investigation and tracking.

Playbook Step	Status	Notes
Alert Detection	Success	Suspicious HTTP activity detected
Check IP Reputation	Success	IP flagged as malicious
Block IP	Success	192.168.1.102 blocked via simulation
Create Case	Success	Incident logged in TheHive

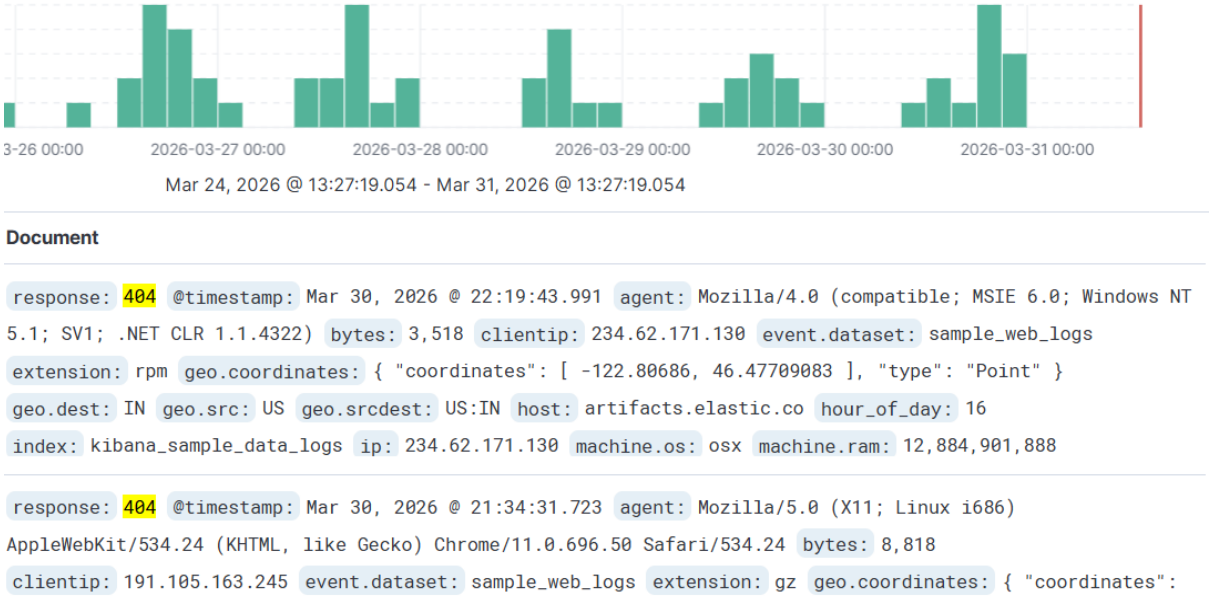

Figure 1: Suspicious HTTP activity detected in SIEM (Elastic/Kibana) showing multiple 404 and 500 responses.

Description 1:
The SIEM system detected abnormal web traffic patterns, including repeated HTTP 404 and 500 errors. Such behavior may indicate phishing attempts, scanning activity, or malicious access attempts. This alert triggered the SOAR playbook for automated incident response.

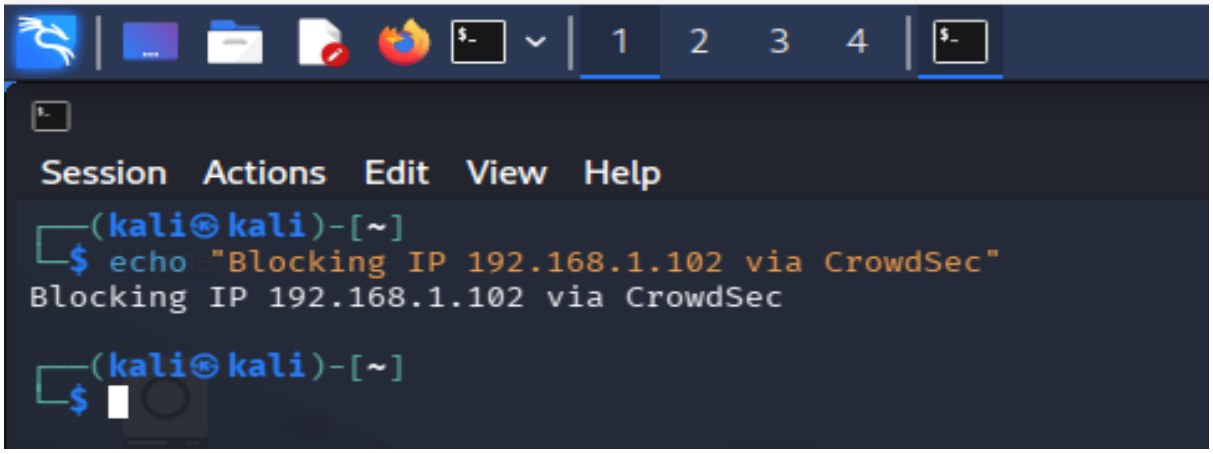

Figure 2: Malicious IP 192.168.1.102 blocked using simulated CrowdSec command in Kali Linux.

Description 2:
As part of the SOAR playbook, the identified malicious IP address was blocked. Due to lab limitations, the blocking action was simulated using a command in Kali Linux to represent automated response via CrowdSec. This step prevents further communication from the attacker.

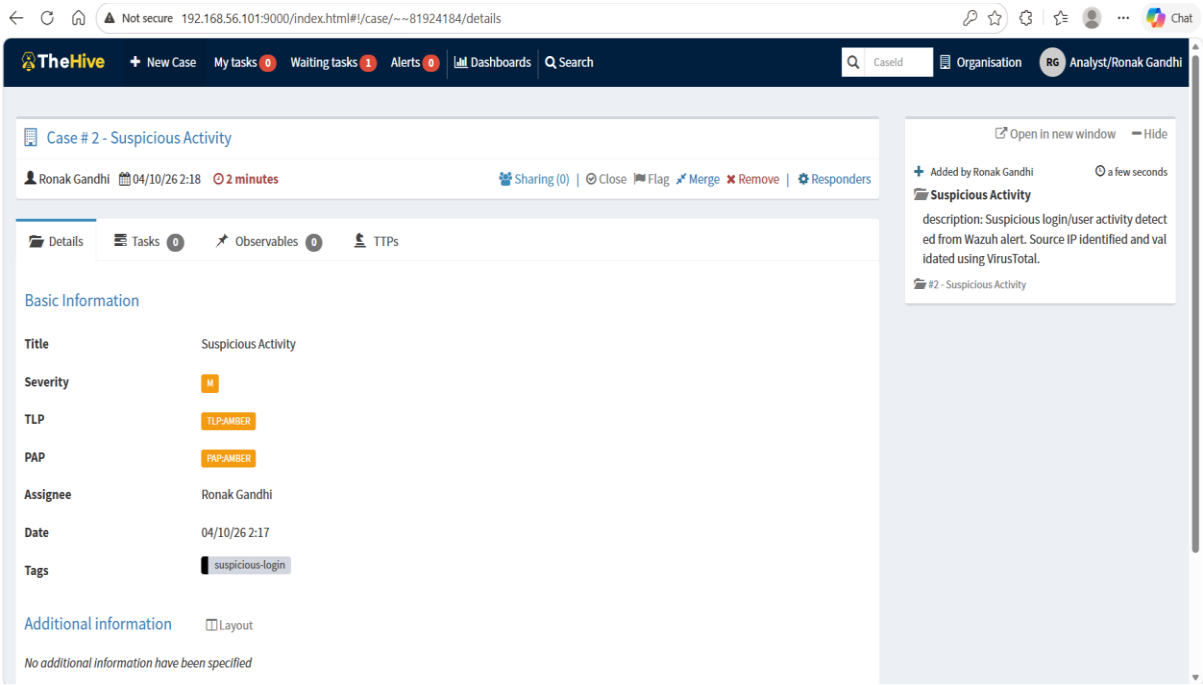

Figure 3: Incident case created in TheHive platform showing suspicious activity detected from SIEM alert.

Description 3:
After detecting and responding to the phishing alert, an incident case was created in TheHive. The case includes important details such as severity, tags, and analyst assignment, allowing proper tracking, investigation, and management of the security incident.

EXPLANATION

The SOAR playbook successfully automated the phishing incident response. Suspicious activity was detected, the malicious IP was identified and blocked, and an incident case was created. Some actions were simulated due to lab limitations, but the workflow demonstrates effective automation.

50-WORD SUMMARY

This playbook automates phishing incident response by detecting suspicious activity, checking IP reputation, blocking malicious IPs, and creating incident cases. It improves response time and enhances security operations using SIEM, CrowdSec, and TheHive integration.